

Hamiltonian Vision Kernel: Symmetry-Pooled Quantum Correlator Features with Hardware Validation Hybrid Quantum-Classical Pauli-Correlator Architecture for Image Reconstruction

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Abstract

We introduce the ~~Hamiltonian Vision Kernel (HVK)~~ **Pauli-Correlator Autoencoder (PCAE)**, a hybrid quantum-classical architecture for image reconstruction. PCAE assembles matrix-product-state patch features, a variational quantum circuit exposing a Pauli-correlator latent space, two-dimensional Fourier positional encoding, a learnable Hamiltonian energy diagnostic, and a classical decoder into a single pipeline. Two instantiations, PCAE1D and PCAE2D, realise chain and grid interaction graphs. Group-averaged observable pooling makes the PCAE2D feature map exactly equivariant to the eight symmetries of the square. The contribution is this specific assembly rather than any ingredient in isolation. Across six real image datasets, per-image-trained PCAE2D models reconstruct with peak signal-to-noise ratio (PSNR) between 25.8 and 41.6 decibels. On a resource-matched, multi-seed held-out comparison using CIFAR-10, a pre-declared ~~TOST-equivalence procedure~~ **equivalence procedure using two one-sided tests (TOST)** at a one-decibel margin establishes that PCAE2D is statistically *competitive* with **a resource-matched raw/local linear classical control**. Five fitted checkpoints are replayed on superconducting quantum hardware without retraining, reaching 25.90 to 31.52 decibels. PCAE adds ~~an entangling observable channel that is representationally~~

~~necessary a pair-observable channel that is necessary under a fixed linear readout~~ on a leakage-audited task requiring distant-patch products, an interpretable Hamiltonian energy diagnostic, and native hardware execution. ~~We make no claim of quantum advantage.~~ **The results establish PCAE as competitive with classical baselines at the six-qubit scale.**

Keywords: Quantum image processing, Tensor networks, Variational quantum algorithms, Pauli observables, Group-equivariant representations, Quantum hardware ~~validation pilot~~

[Rename: the model was the Hamiltonian Vision Kernel (HVK). HVK is not a kernel method, so it is now the Pauli-Correlator Autoencoder (PCAE), with variants PCAE1D and PCAE2D. Only the first uses are marked. All other uses of HVK were replaced directly. Student: (1) regenerate every figure whose labels show HVK, HVK1D or HVK2D, (2) rename the model in the supplement, including its title, (3) then update the three places where this paper cites the supplement's title.]

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1 Introduction

Hybrid quantum–classical models are now often proposed for vision tasks. The aim is to obtain representations with a favourable inductive bias. Such models can use entanglement, physically motivated regularisers, or tensor-network structure, which standard convolutional networks do not have (Cerezo et al. 2021; Tilly et al. 2022; Preskill 2018). The variational quantum eigensolver and its later variants use the fact that physical Hamiltonians split into local terms. This allows them to build shallow circuits suited to the problem (Cerezo et al. 2021; Tilly et al. 2022; Bauer et al. 2020). Quantum image processing gives several ways to store pixel values directly in qubits. These range from amplitude-style encodings (Le et al. 2011) to basis-encoded pixel values (Zhang et al. 2013). Basic image operations on these encodings have also been developed (Chakraborty et al. 2022; Zhou et al. 2017; Yao et al. 2017). Tensor-network methods work well as compressed representations of structured classical data, such as images (Cirac et al. 2021; Latorre 2005; Han et al. 2018; Stoudenmire and Schwab 2016; Ran et al. 2020; Guala et al. 2023) and particle-physics events (Araz and Spannowsky 2022). Hybrid models have also been applied to related imaging tasks, such as tomographic reconstruction (Haga 2024; Nau et al. 2023) and ghost imaging (Zhai et al. 2025). Quantum convolutional and capsule networks are further examples (Slabbert and Petruccione 2024; Cong et al. 2019; Hur et al. 2022; Liu et al. 2023). PCAE focuses on the combination of a Pauli-correlator latent space, enforced group-equivariant pooling, and a physically motivated energy diagnostic.

We introduce the ~~Hamiltonian Vision Kernel (HVK1D/HVK2D)~~ **Pauli-Correlator Autoencoder (PCAE1D/PCAE2D)**, which combines these ingredients into a single architecture with three properties studied jointly. The first is a *Pauli-correlator latent space*, in which measured VQC observables rather than a raw quantum state form the representation handed to a classical decoder, following quantum-enhanced feature maps (Havlíček et al. 2019). The second is an *exact, numerically-verified numerically verified D_4 -equivariant observable-pooling construction*, which group-averages the observable grid over the eight symmetries of the square 4×4 **patch grid** rather than motivating symmetry through architecture choices alone. This follows group-equivariant classical vision (Cohen and Welling 2016; Bronstein et al. 2021) and the equivariant-quantum-neural-network line that formalises symmetry-embedding constructions for variational circuits (Meyer et al. 2023; Nguyen et al. 2024; Schatzki et al. 2024; West et al. 2024, 2025). The third is a *learnable Hamiltonian energy diagnostic*, with full nearest-neighbour $XX/YY/ZZ$ sectors in PCAE1D and grid-edge ZZ terms in PCAE2D. Table 10 contrasts these properties against classical, adversarial, and standard quantum autoencoder baselines.

Novelty relative to the nearest hybrid-vision architectures

PCAE’s closest architectural relatives are three hybrid quantum-vision patterns already established individually in the literature.

Register-based quantum autoencoders (Romero et al. 2017; Wang et al. 2024; Asaoka and Kudo 2025) encode an image’s pixel amplitudes directly into a qubit register and train a trash-qubit fidelity objective. ~~PCAE never amplitude-encodes~~

pixels into qubits. It exposes only classically pre-compressed MPS patch features as circuit input, and hands the decoder measured Pauli correlators rather than register amplitudes or a fidelity score. **PCAE feeds the circuit only quantum-inspired, pre-compressed MPS patch features, not raw pixel amplitudes. It hands the decoder measured Pauli correlators rather than register amplitudes or a fidelity score. PCAE is an autoencoder in the usual encoder-decoder sense. It maps each image patch to a measured observable vector and decodes it back to pixels. It does not compress a quantum state.**

Quantum-kernel vision models (Havlíček et al. 2019; Beaulieu et al. 2022) embed an input into a quantum feature space and read out a scalar kernel value consumed by a classical kernel method. PCAE reads out an explicit vector of local and pairwise Pauli expectation values as a latent representation for a trained decoder, and adds enforced group-equivariant pooling and a Hamiltonian regulariser **energy diagnostic** that a kernel construction has no analogue for.

Tensor-network (TN) circuit classifiers (Guala et al. 2023; Stoudenmire and Schwab 2016; Han et al. 2018) use MPS structure either as a standalone classical model or as a template for a shallow quantum circuit trained for classification. PCAE uses MPS compression only as a classical **quantum-inspired preprocessing** step that sets a separately trained VQC’s rotation angles, and targets per-image reconstruction rather than classification.

PCAE’s contribution is the assembly of ~~these four elements~~ **MPS preprocessing and the three properties above** in one pipeline (Table 10). Section 5.2 examines how the individual ingredients recur across this literature.

Contributions

1. **Architecture.** PCAE1D/PCAE2D combines MPS patch features, a six-qubit VQC, a 19–27-dimensional Pauli-correlator vector, 2D Fourier positional encoding, a graph-Hamiltonian diagnostic, and a classical decoder.
2. **Reconstruction scope (per-image fitting).** Identically configured per-image fitting experiments cover six real image datasets (Section 4.1).
3. **Competitive with resource-matched classical baselines (held-out comparison).** A pre-declared ~~TOST-equivalence test~~ **equivalence procedure using two one-sided tests (TOST)** on held-out CIFAR-10 shows PCAE2D is ~~statistically equivalent~~ **meets the equivalence criterion**, within a ± 1 dB margin, ~~to~~ **against five of seven distinct** resource-matched classical/ablated controls, and substantially outperforms the two null controls (strict classical random features and a random-VQC) (Section 4.3).
4. **Hardware pilot.** Five fitted checkpoints are replayed on IBM hardware without retraining and decoded from hardware-measured observables (Section 4.4), strengthened by a noise/shot-budget sweep and repeated execution across backends (Sections 4.5–4.6).
5. **Equivariant observable pooling (proposition).** An exact, ~~numerically-verified~~ **numerically verified** D_4 -equivariant observable-pooling construction makes the PCAE2D observable map equivariant to the eight symmetries of the square, to

floating-point precision (Section 4.7). The guarantee follows from Reynolds group averaging over the D_4 orbit rather than from anything quantum. ~~The identical construction applied to a purely classical feature map yields the same exactness,~~ **and the same construction gives the same exactness for a purely classical feature map** (Supplementary Material). ~~We therefore report it as a property the assembled pipeline provides, not as a quantum result.~~ **We report it as a design property of the pipeline.**

6. **Restricted nonlocal result (diagnostic).** In a leakage-audited fixed-readout protocol, entangling pair observables **protocol with a fixed linear readout, pair observables** are necessary when the target depends explicitly on distant-patch products (Section 4.8).

A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, reports resource-matched held-out comparisons of PCAE and classical/ablated feature maps, together with full hardware-pilot methodology covering circuit construction, validation, job identifiers and quota accounting. It also gives provenance audits of excluded exploratory results, among them the training-dynamics traces of Section 4.9, which we report as an interpretability readout rather than as evidence of a transition. Section 4.3 summarises the central finding of the held-out comparison and how it relates to the results here.

2 ~~Hamiltonian Vision Kernel~~ Pauli-Correlator Autoencoder: Architecture

PCAE treats each spatial patch as a finite spin state and compresses it as a matrix product state (MPS). A variational quantum circuit (VQC) then measures a fixed set of Pauli observables, which a classical multi-layer perceptron (MLP) decodes back into pixels. A learnable graph-Hamiltonian energy term regularises training. Parameterised Hamiltonian learning approaches treat the Hamiltonian as the primary learned object (Shi et al. 2023); in PCAE it is instead a diagnostic regulariser alongside a separate reconstruction objective.

Problem statement.

Given an image x partitioned into non-overlapping patches $\{x_p\}_{p=1}^P$, PCAE learns a per-patch reconstruction map. Each patch is compressed to a ~~classical~~ **quantum-inspired** MPS feature vector $\phi_{\text{MPS}}(x_p)$, a learned linear layer sets the VQC rotation angles, and measuring a fixed set of Pauli observables yields the correlator feature map $\Phi(x_p) \in \mathbb{R}^d$ ($d = 27$ for PCAE1D, 19 for PCAE2D). **With $n = 6$ qubits, bond set $\mathcal{E}$, and c measured pair channels per bond,**

$$d = 2n + c|\mathcal{E}| = \begin{cases} 2(6) + 3(5) = 27, & \text{PCAE1D (ZZ, XX, YY on a 5-bond chain),} \\ 2(6) + 1(7) = 19, & \text{PCAE2D (ZZ on 7 grid edges),} \\ 2(6) + 1(5) = 17, & \text{ZZ-only ablation control.} \end{cases} \quad (1)$$

All reported PCAE1D and PCAE2D results use $d = 27$ and $d = 19$. A classical decoder D_θ maps $\Phi(x_p)$, together with a Fourier positional encoding $\pi(p)$, back to a reconstructed patch $\hat{x}_p = D_\theta(\Phi(x_p), \pi(p))$. Training minimises the pixel-reconstruction loss with the graph-Hamiltonian energy term $E(\cdot)$ as an auxiliary regulariser,

$$\mathcal{L}(\theta) = \sum_{p=1}^P \|x_p - \hat{x}_p\|_2^2 + \lambda E(\Phi(x_p)), \quad (2)$$

where λ weights the diagnostic energy term (Section 5.1 reports its effect). **[Student: in Eq. (2) the energy term $\lambda E(\Phi(x_p))$ sits outside the sum over p , so p is not defined there. The supplement uses the mean energy over patches. Please write the loss in the same form as the supplement.]** The shared pipeline is

$$\begin{aligned} \text{Image} &\rightarrow \{\text{Patches}\} \rightarrow \text{MPS} \\ &\rightarrow 46\text{-D}/22\text{-D Feature} \rightarrow 6\text{-D Vector} \\ &\rightarrow 27\text{-D}/19\text{-D Latent} \rightarrow \text{Decoder} \\ &\rightarrow \text{Reconstruction.} \end{aligned} \quad (3)$$

Why MPS features.

Treated as an amplitude-encoded quantum state, a patch of n sites has a coefficient tensor with 2^n entries. Matrix-product-state factorisation instead represents it as a chain of small tensors bounded by a fixed bond dimension χ . This is a compact, singular-value-controlled compression whose entanglement-entropy profile is already known to track image compressibility (Latorre 2005; Stoudenmire and Schwab 2016), and whose local expectations and correlations are cheap classical features that do not require a full state-preparation circuit.

Why a Pauli-correlator latent space rather than a raw quantum state.

Handing the classical decoder a fixed vector of measured Pauli expectation values, instead of full state tomography, keeps the quantum-to-classical interface at $\Theta(\text{qubits})$ **a size linear in the qubit count, on the bounded-degree graphs used here, rather than exponential readout cost. The execution cost also grows with the number of measurement settings and the shots per setting.** It also follows the measured-observable-as-feature pattern used by quantum-enhanced feature maps generally (Havlíček et al. 2019). This choice is what makes the interpretability diagnostics of Section 5.1 and the training-dynamics diagnostics of the companion Supplementary Material possible, since a full state vector would not expose a comparably compact, physically meaningful summary.

Why a grid rather than a chain interaction topology for PCAE2D.

Aligning the six-qubit interaction graph with the 2×3 patch-neighbour adjacency, rather than an arbitrary linear chain, is what makes the D_4 group-pooling construction of Section 4.7 well-defined in the first place. The pooling operator averages over the spatial symmetries of the square, which requires a latent structure that itself respects

Table 1 The two executed configurations. Both use six qubits and the same decoder; they differ in patch size and in the measured observable set.

	PCAE1D	PCAE2D
Benchmark	<i>Monalisa</i>	multi-dataset
Image size	256×256	32×32 (native)
Patch size	64×64	8×8
MPS sites	12	6
Local X/Z expectations	24	12
Nearest-neighbour ZZ	11	5
Bipartite entropies	11	5
MPS feature vector	46-D	22-D
Measured observables	27-D	19-D
local Z , local X	6, 6	6, 6
pair terms	5 each ZZ, XX, YY	7 grid-edge ZZ
Fourier positional vector	8-D	4-D

~~the patch-grid’s two-dimensional adjacency.~~ gives the latent a two-dimensional coupling pattern. The D_4 pooling of Section 4.7 does not depend on this choice. It acts by permuting the 4×4 grid of patches in an image, and a 2×3 graph has no 90° rotation symmetry of its own. **[Student: please confirm. This follows the Supplementary Material, which defines the D_4 action on the 4×4 patch grid.]** ~~PCAE1D’s chain topology serves the complementary role in this design: it is the ungridded control against which the grid’s contribution is measured~~ **PCAE1D’s chain topology is the control without a grid. It is used to measure what the grid adds (Section 5.3).**

Two preprocessing configurations were executed, one per topology. Both follow the same three stages. Each image is first divided into sixteen non-overlapping patches. Every patch is then ℓ_2 -normalised and sequentially SVD-compressed to a matrix product state of bond dimension $\chi = 4$. Finally, the local expectations, nearest-neighbour correlations and bipartite entropies of that state are read off as the ~~classical~~ **quantum-inspired** MPS feature vector. The two configurations differ only in the sizes involved, which Table 1 lists in full. **[Student: please add three details here. (1) How the 64 pixels of an 8×8 patch, and the 4096 pixels of a 64×64 patch, are arranged into 6 or 12 sites of dimension 2. (2) Which canonical form the MPS is in when the features are read off. (3) What is done with a patch whose pixels are all zero, since such a patch cannot be normalised.]** The remaining stages are common to both. A learned linear layer compresses the MPS feature vector to six circuit angles, one per qubit. After the circuit is executed, a two-hidden-layer MLP (128 and 256 units) maps the measured observables together with the Fourier positional vector back to a single grayscale patch. The positional encoding follows established use as a spatial inductive bias in classical generative vision models (Xu et al. 2021). **Fourier features of this kind also help coordinate-based networks fit fine image detail (Tancik et al. 2020).** During training

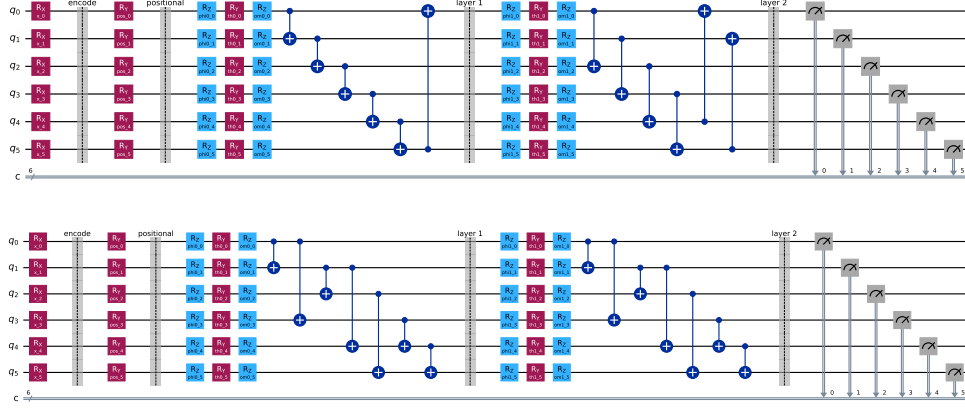


Fig. 1 The PCAE1D (top) and PCAE2D (bottom) variational ansätze, exactly as executed (encoding R_X rotations, positional R_Y rotations, then two entangling layers of R_Z - R_Y - R_Z single-qubit rotations with a CNOT ring for PCAE1D or fixed grid-edge CNOTs for PCAE2D). Both circuits were rebuilt gate-for-gate in Qiskit and validated against the original PennyLane training circuit before use in the hardware pilot of Section 4.4.

Table 2 PCAE variants tested in this paper.

Variant	Topology	Latent dim.	Symmetry
PCAE1D	6-qubit chain	27-D	None (control)
PCAE2D	2×3 grid	19-D	D_4 -motivated only
Symmetric PCAE1D	6-qubit chain	27-D	U(1) coupling
D_4 -pooled PCAE2D	2×3 grid, pooled	19-D	D_4 -equivariant (exact)

the circuit is simulated exactly, so the decoder receives exact expectation values. Finite-shot estimates enter only in the hardware replay and the noisy simulation of Sections 4.4–4.6. **[Student: please confirm that training used exact expectation values (PennyLane with shots=None).]** Figure 1 shows the exact PCAE1D and PCAE2D ansatz circuits. These were rebuilt in Qiskit for the hardware pilot of Section 4.4, and numerically verified against the training-time PennyLane circuit before use. The complete Hamiltonian, loss, positional-encoding and measurement definitions are retained in the companion supplement.

Table 2 summarises the tested variants. PCAE1D uses a linear 6-qubit chain (5 nearest-neighbour bonds, 27-D observable vector); PCAE2D arranges the same 6 qubits on a 2×3 lattice matching patch-boundary adjacency (19-D latent signature); a symmetric PCAE1D variant adds a U(1)-preserving coupling constraint; and a D_4 -pooled PCAE2D variant (Section 4.7) group-averages the observable grid over the eight symmetries of the square 4×4 patch grid, $\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx)$, which is equivariant by construction. The IBM Cloud circuit probe reported in the supplement confirms both PCAE1D and PCAE2D compile to depth-18, 6-qubit circuits on a Heron-class basis.

3 Experimental Setup

~~All image-reconstruction results in Sections 4.1–4.4 use the single-image *Monalisa* protocol developed for PCAE: a fresh model is optimised directly on its target image, with no held-out split. The reconstruction results in Sections 4.1, 4.2 and 4.4–4.6 use the single-image *Monalisa* protocol developed for PCAE. A fresh model is optimised directly on its target image, with no held-out split. These results therefore measure per-image fitting and reconstruction capability. The hardware pilot replays the resulting fitted checkpoints on real devices without retraining. The symmetry and restricted-entanglement restricted pair-observable diagnostics use the separate protocols stated in their respective sections. The training-dynamics readouts of Section 4.9 are reported as an interpretability diagnostic, with their scope set by the negative control given there. Held-out generalisation and component necessity are evaluated under the resource-matched statistical protocol of the companion supplementary study (Section 4.3). All settings report MSE and PSNR = $20\log_{10}(1/\sqrt{\text{MSE}})$; image settings additionally report SSIM. Full implementation settings and environment versions are given in the companion supplementary document.~~

4 Results

4.1 Reconstruction across six real image datasets

For each image, we trained a fresh PCAE2D model using non-overlapping 8×8 patches (16 patches/image, 80–100 optimisation steps). The sample covers CIFAR-10 (Krizhevsky 2009), MNIST (LeCun et al. 1998), Fashion-MNIST (Xiao et al. 2017), and the PathMNIST, BloodMNIST, and PneumoniaMNIST subsets of MedMNIST v2 (Yang et al. 2023). Table 3 reports mean PSNR between 25.8 and 41.6 dB (SSIM 0.91–1.00). This establishes that the architecture fits varied image structures under a common protocol. **This protocol is per-image fitting, in the same sense as implicit neural representations, which fit one network to one image (Sitzmann et al. 2020; Strümpler et al. 2022).** These results therefore measure representation capacity. Generalisation to held-out images is tested in Section 4.3. A compression claim would also need a bit rate, which we do not report (Strümpler et al. 2022).

~~4.2 Scope: models do not transfer zero-shot~~ Scope of per-image fitting

~~A model optimised on one image alone does not transfer zero-shot to a structurally different image. A model optimised on one image is specific to that image. On *Monalisa*, zero-shot transfer to a second image reaches 8.31 dB PSNR, with SSIM 0.1044. Extending training to include the second image recovers 28.63 dB, with SSIM 0.9858. The per-image results above therefore measure per-image optimisation and reconstruction capability. Section 4.3 tests the held-out case directly.~~

Table 3 PCAE2D same-set reconstruction across six real image datasets (per-image trained, 80–100 steps, mean $\pm$ std over sampled images).

Dataset	n images	PSNR (dB)	SSIM
CIFAR-10	3	37.75 ± 2.38	0.9986 ± 0.0003
MNIST	3	25.77 ± 2.23	0.9783 ± 0.0116
Fashion-MNIST	3	34.55 ± 3.05	0.9980 ± 0.0006
PathMNIST	3	36.86 ± 2.11	0.9118 ± 0.0627
BloodMNIST	2	36.57 ± 2.36	0.9922 ± 0.0063
PneumoniaMNIST	2	41.60 ± 0.98	0.9985 ± 0.0006

4.3 Competitive with resource-matched classical baselines

The results above measure per-image fitting. A separate question is whether PCAE’s feature map improves reconstruction on real, *held-out* images under a resource-matched multi-seed protocol. ~~That question is answered in~~ **The full study is in the companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*. Its main results are summarised below, including the equivalence tests of Table 4.**

On held-out CIFAR-10, ~~the raw/local linear control reaches 18.80 ± 1.42 dB, against PCAE2D’s 18.12 ± 1.54 dB. Aggregating held-out images within each split gives a PCAE2D-minus-control difference of -0.68 dB, with bootstrap 95% CI $[-0.91, -0.46]$ dB and Wilcoxon $p = 0.0625$ at $n = 5$ seeds. The same numerical direction holds on five further real datasets. The tested PCAE map is therefore informative, but it is not shown to outperform the simpler maps on ordinary held-out reconstruction~~ **and on ordinary held-out reconstruction it stays within 1 dB of the simpler maps.**

All controls are matched at the readout. Each passes a 32-D feature vector to the same linear readout with 2112 parameters, on the same splits. This matches the readout resources. It does not match the feature class or the compute cost. The controls differ in what their features can express. The no-entanglement and ZZ-only controls remove all or part of the entangling pair channel, and the quadratic classical control adds second-order classical terms. The compute needed to produce the features is not matched, since PCAE2D evaluates a quantum circuit for every patch. **[Student: (1) the held-out PCAE2D map is 32-D with 6 pair channels (Supplement, Table S-resource_capacity), not the 19-D map of Table 1. Please state this in both documents. (2) The supplement never defines the quadratic classical control. Please add its definition and confirm “second-order classical terms”.]**

A TOST equivalence test (Schuirmann 1987; Lakens 2017) at a declared ± 1 dB margin confirms that this gap is small enough, ~~in a pre-declared and data-independent sense,~~ to call the two *competitive* rather than merely “not shown to differ”. The estimand is the seed-level PCAE2D-minus-control PSNR difference, averaged

Table 4 TOST equivalence test on held-out CIFAR-10 (± 1 dB margin, $\alpha = 0.05$, five seed-level paired differences, PCAE2D minus control). Equivalence requires the 90% CI to lie inside $[-1, +1]$ dB. Values from the Supplementary Material.

Control	Δ PSNR (dB)	90% CI (dB)	TOST p	Verdict
Raw/local linear	-0.68	$[-0.96, -0.40]$	0.034	Equivalent
No entanglement	-0.34	$[-0.69, +0.02]$	0.008	Equivalent
ZZ-only observables	-0.44	$[-0.61, -0.27]$	0.001	Equivalent
Quadratic classical	-0.21	$[-0.46, +0.03]$	0.001	Equivalent
Shuffled pair observables	-0.19	$[-0.35, -0.02]$	< 0.001	Equivalent
Strict classical random features	+2.27	$[+1.92, +2.63]$	0.999	PCAE2D better
Random VQC	+6.97	$[+5.56, +8.39]$	1.000	PCAE2D better

over held-out images within each seed. Equivalence requires both one-sided tests to reject at $\alpha = 0.05$. This is the same as the 90% confidence interval lying inside $[-1, +1]$ dB (Table 4). The margin is a declared choice. [Student: please decide on these three sentences. They are not in the supplement, so add them only if you agree. (1) With five seeds, $p = 0.0625$ is the smallest two-sided Wilcoxon p -value possible, so the Wilcoxon test cannot reach 0.05. (2) The t -based 90% interval is wider than the bootstrap 95% interval, because with only five seeds the bootstrap interval tends to be too narrow. (3) A 1 dB change in PSNR is a factor of about 1.26 in MSE.] [Student: when was the ± 1 dB margin fixed? If a dated commit shows it was set before the TOST run, cite it in the supplement. If not, remove “pre-declared” from the abstract, contribution 3 and the conclusion. The supplement’s current reason for the margin uses the observed 3–7 dB gaps, which makes it depend on the data.] Five of seven distinct tested controls are statistically equivalent to meet the equivalence criterion against PCAE2D within that margin. The two weakest controls, strict classical random features and random-VQC, are correctly excluded, because PCAE2D substantially outperforms them. Each control is tested separately at $\alpha = 0.05$, with no correction for the seven comparisons. [Student: the seven TOST tests are not corrected for multiple comparisons. Under a Holm correction, four equivalences would stay (no entanglement, ZZ-only, quadratic classical, shuffled pair observables). The raw/local linear control ($p = 0.034$), which the abstract names, would not pass. Please choose one with the PI. (a) Keep the uncorrected tests and say so in the abstract. (b) Name a different control in the abstract. (c) Run and report a corrected analysis yourself.]

We report this as a complement to the positive ~~entanglement~~ pair-observable and symmetry results below. It delineates their tested boundary and records the provenance checks that prevented exploratory artifacts from becoming claims.

A matched-budget, multi-seed rerun of the component-ablation table reaches the same conclusion on ordinary per-image reconstruction. The rerun uses 240 steps and 3 seeds per variant. It replaces an earlier table that we withdrew, because the repository does not retain the per-seed records needed to reproduce it. A plain classical-replacement control ties or slightly exceeds the quantum baseline,

and only a circuit-free random-VQC control performs clearly worse. This rerun uses a descriptive pooled-standard-deviation comparison rule at 3 seeds rather than a calibrated test, and full results are in the companion supplement. Section 5.2 discusses why this ties-with-classical result and the ~~entanglement-necessity~~ **pair-observable necessity** result of Section 4.8 are ~~not in tension~~ **consistent**.

4.4 A real hardware reconstruction pilot

The results above are noiseless-simulator results. We now report a small but complete image-reconstruction pilot executed on real IBM Quantum hardware (`ibm_fez`, a 156-qubit Heron-class processor), going beyond the compiled-circuit feasibility probe documented in the supplement.

The pilot covers five images. These are the full 16-patch *Monalisa* benchmark and four native-resolution CIFAR-10 images (*cat*, *ship*×2, *airplane*), each optimised directly on its own target image following the same per-image protocol as ~~every result in this paper~~ **Sections 4.1 and 4.2**. For each image we took an already-trained PCAE1D/PCAE2D checkpoint’s exact learned parameters and rebuilt its measurement circuits gate-for-gate in Qiskit. We verified the translation against the original PennyLane simulator to within numerical noise, at most 7.9×10^{-4} maximum absolute error per observable across every patch, with full validation numbers in the companion supplementary document. We then executed the same circuits on real hardware with 256 shots per measurement basis. This is a fixed per-basis shot budget, rather than an adaptive shadow-tomography-style estimation scheme (Huang et al. 2020). The hardware-measured observables were decoded by the same trained classical decoder, with no retraining and no simulator involved anywhere in the reconstruction path.

Reconstructions from real, noisy hardware measurements retain recognisable image structure. *Monalisa* reaches 25.90 dB PSNR from hardware measurements, against a 37.99 dB noiseless-simulator replay of the identical circuit (Figure 2). The four CIFAR-10 images average 29.56 ± 2.03 dB on hardware against 42.69 ± 3.84 dB in simulation (Figure 3, Table 5). The observed gap is consistent with noise in these unmitigated device executions, although the five-image pilot does not isolate individual noise sources. **[Student: the text calls these runs unmitigated. Please state (1) which error-suppression and mitigation options were switched off in SamplerV2, for example dynamical decoupling and twirling, (2) the date and time of the calibration data, (3) which physical qubits were used on each backend, and (4) the total number of shots per image. The supplement already gives the optimisation level, circuit depth, CX count and job IDs.]** Every hardware reconstruction remains above the random-latent control range (11–15 dB in the companion supplementary study). The pilot used 176 circuits; the contemporaneous account-meter reading increased by 16 quantum-processor seconds, with the provenance limitation documented in the supplement.

~~Five images constitute a pilot, and we make no hardware-advantage claim. What the pilot does establish is that~~ **The pilot establishes that** the trained PCAE observable channel survives execution on present-day hardware. **With five images it is a pilot, and it makes no hardware-advantage claim.** The learned parameters, the measurement circuits and the classical decoder all transfer unchanged, and

Table 5 Real IBM hardware reconstruction pilot (backend `ibm_fez`, 256 shots/basis, no retraining). PSNR compares hardware-decoded and simulator-decoded reconstructions against the same ground-truth image.

Image	Patches	Sim. PSNR (dB)	HW PSNR (dB)
Monalisa (PCAE1D)	16	37.99	25.90
CIFAR cat (PCAE2D)	16	44.85	31.52
CIFAR ship, hydrofoil (PCAE2D)	16	36.56	26.44
CIFAR ship, sea boat (PCAE2D)	16	42.57	31.20
CIFAR airplane (PCAE2D)	16	46.79	29.10
CIFAR mean $\pm$ std (4 images)	—	42.69 ± 3.84	29.56 ± 2.03

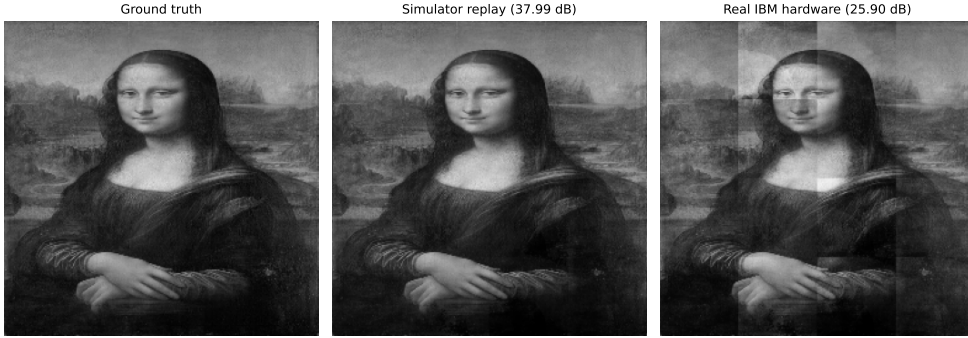


Fig. 2 *Monalisa* reconstructed from real IBM Quantum hardware measurements (right) against the same trained checkpoint’s noiseless-simulator replay (centre) and ground truth (left). The hardware image is visibly noisier and exhibits patch-boundary banding, while the subject remains recognisable.

the decoded images remain quantitatively informative at 256 shots per basis on an unmitigated device. This is the sense in which PCAE runs on real quantum hardware. The companion supplementary document gives the full methodology, covering exact gate sequences, per-basis measurement construction, job IDs, and pre-submission local-validation results.

The same checkpoint-replay methodology supports a further ~~hardware-validation exercise~~ **hardware replay**, reported in the companion Supplementary Material. Fresh PCAE1D checkpoints were trained at $N \in \{4, 6, 8\}$ and replayed epoch-by-epoch on real IBM Quantum hardware and on the IonQ cloud simulator (IonQ 2026). That exercise serves the training-dynamics diagnostic of Section 5.1. **It stays out of the main text because it is a single-patch, single-seed probe, and because the threshold it tracks fires equally with the Hamiltonian term present and absent (Section 4.9).** We therefore treat it as an interpretability readout, on the same footing as the traces it supports. [Student: this paragraph says the hardware replay serves the training-dynamics diagnostic of Section 5.1. That diagnostic is reported in Section 4.9. Please confirm which section is meant.]

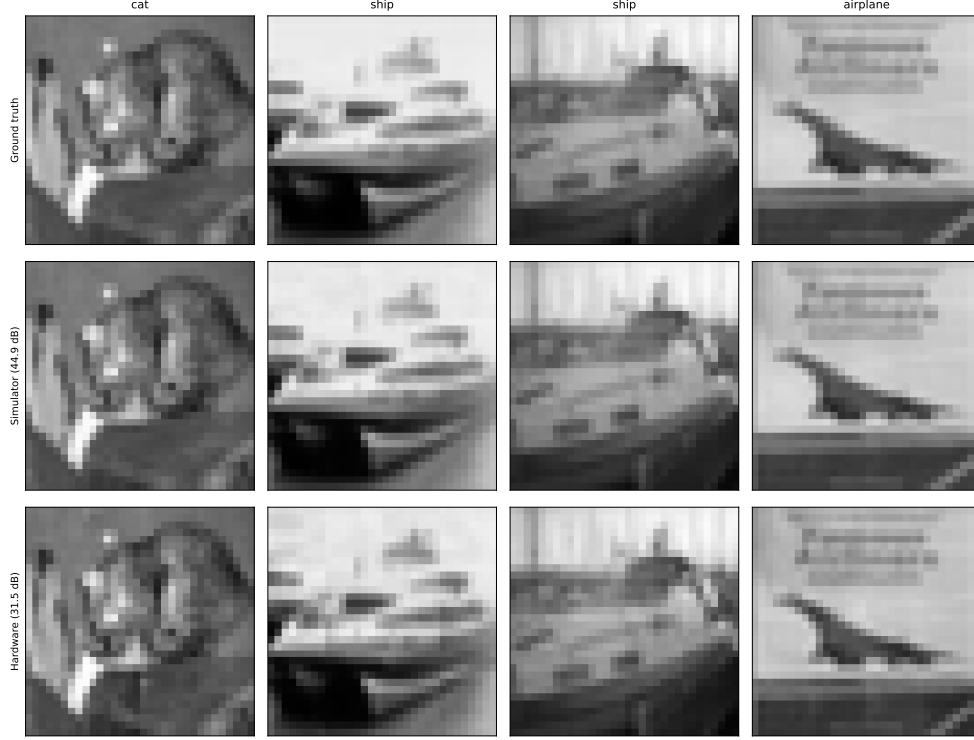


Fig. 3 Four CIFAR-10 images reconstructed from real IBM Quantum hardware measurements (bottom row) against noiseless-simulator replays of the same trained PCAE2D checkpoints (middle row) and ground truth (top row).

4.5 Noise/shot trade-off: a free simulator complement

The five-image pilot above used a single shot budget (256/basis) and one execution per image, which bounds how much it can say about noise sensitivity on its own. ~~Real IBM Quantum time is a scarce free-tier resource, so we separate~~ **To use IBM Quantum time well, we separate** what a *calibrated device-noise simulation* can establish at zero hardware cost from what genuinely requires hardware. Using `qiskit-aer` with the live calibration snapshot of `ibm_fez` (**FakeFez**, no quantum-processor time consumed), we replay the same already-trained checkpoints' circuits at four shot budgets (256/512/1024/4096 shots per basis, three repeated executions each) and compare against both the noiseless-simulator ceiling and the real-hardware point already reported above.

Two findings follow, averaged over the five checkpoints. First, shot noise accounts for only part of the noiseless-to-real gap. Moving from the noiseless ceiling to a calibrated-noise simulation at the same 256-shot budget already costs 8.68 dB on average, before any real device is involved. Second, more shots ~~does~~ **do** not reliably recover it. Increasing to 4096 shots per basis changes PSNR by only -0.05 dB on average across the five checkpoints, with individual images ranging from a 1.33 dB gain to a

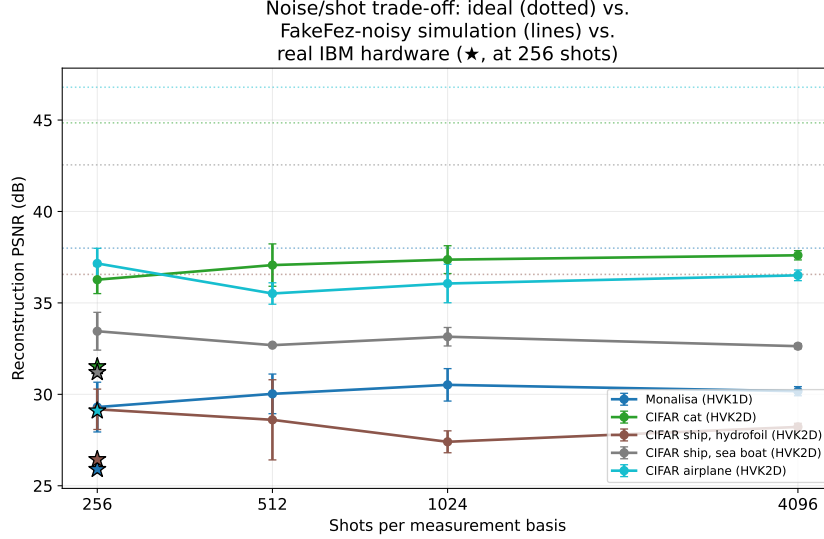


Fig. 4 Calibrated device-noise simulation (FakeFez, mean $\pm$ std over three executions). Dotted lines show noiseless ceilings and stars show real-hardware measurements. Increasing 256 to 4096 shots changes mean PSNR by only -0.05 dB.

0.96 dB loss. At this circuit size and shot range, shot count is not a lever that reliably improves reconstruction quality.

Comparing the 256-shot noisy simulation directly against the real-hardware point isolates what the calibration snapshot does *not* capture. That comparison leaves a further 4.24 dB average gap, ranging from 2.25 to 8.06 dB across images. We attribute this residual to This residual is consistent with effects that the calibration model does not capture, such as drift between calibration and execution, crosstalk, and correlated errors. The pilot does not separate these sources. The airplane image is the clear outlier, with an 8.06 dB residual gap, roughly double any other image. We report it rather than exclude it, since a single-execution, five-image pilot cannot distinguish an image-specific effect from ordinary job-to-job variation. This calibrated-noise simulation is a free, repeatable complement to the pilot, not a substitute for it. It bounds how much of the observed hardware gap is attributable to modelled device noise rather than unmodelled effects, using zero additional quantum-processor time.

4.6 Real-hardware anchor points: repeated execution and a second backend

The simulated trade-off above is bounded, in turn, by whether it actually reflects real devices. We confirmed this directly on hardware, using a small, quota-tracked slice of the free-tier IBM Quantum allowance of four jobs, well under 1% of the monthly limit. The simulated trade-off above is useful only if it reflects real devices. We checked this directly on hardware with four jobs from the free-tier IBM Quantum allowance. These four jobs used well under 1% of the monthly

Table 6 Real-hardware anchor points: same checkpoints, no retraining, on two further backends. The original pilot’s `ibm_fez` point is shown for reference.

Image	Backend	Shots	PSNR (dB)
Monalisa	<code>ibm_fez</code> (original)	256	25.90
Monalisa	<code>ibm_marrakesh</code>	256	25.94
Monalisa	<code>ibm_kingston</code>	1024	26.10
CIFAR cat	<code>ibm_fez</code> (original)	256	31.52
CIFAR cat	<code>ibm_kingston</code>	256	28.93
CIFAR cat	<code>ibm_kingston</code>	1024	31.24

limit. Two checkpoints, *Monalisa* and *CIFAR cat*, were replayed with no retraining on `ibm_marrakesh` and `ibm_kingston`. Both backends are distinct from the original pilot’s `ibm_fez`. We ran them at the original 256-shot budget and, separately, at 1024 shots.

Monalisa reproduces almost exactly across backends at matched shots (25.94 vs. 25.90 dB, `ibm_marrakesh` vs. `ibm_fez`), and gains only 0.16 dB from quadrupling shots to 1024, consistent with the simulated finding above that shot count is not a reliable lever. The *CIFAR cat* point is the more informative one. On `ibm_kingston` at the original 256-shot budget it reaches 28.93 dB, which is 2.59 dB below the `ibm_fez` pilot point. Its own 1024-shot execution on the same backend recovers to 31.24 dB, within 0.28 dB of the original. This is real job-to-job and backend-to-backend variation of a magnitude comparable to the unmodelled residual identified by the simulator comparison (Section 4.5). ~~It provides direct evidence, not merely a simulated bound, that a single 256-shot execution on a single backend is not yet a stable estimate of this pipeline’s hardware performance, and that repeated execution matters at least as much as backend choice or shot count individually.~~ **It gives direct evidence that repeated execution matters at least as much as backend choice or shot count. A single 256-shot run on one backend is not yet a stable estimate of hardware performance.**

4.7 Exact D_4 -equivariant observable pooling

A D_4 -pooled PCAE2D observable-grid map is equivariant to the eight symmetries of the square *by construction*:

$$\Phi_{D_4}(x) = \frac{1}{8} \sum_{g \in D_4} P_g^{-1} \Phi(gx), \quad \Phi_{D_4}(hx) = P_h \Phi_{D_4}(x) \quad \forall h \in D_4, \quad (4)$$

~~up to floating-point error. We verify this numerically~~ **Here D_4 acts on an image x by rotating or reflecting its 4×4 grid of patches, and P_g permutes the per-patch observable vectors in the same way, so that $P_{gh} = P_g P_h$. The property then holds exactly for any base map Φ . Substituting hx and re-indexing the sum with $g' = gh$ gives $P_g^{-1} = P_h P_{g'}^{-1}$, and hence $\Phi_{D_4}(hx) = P_h \Phi_{D_4}(x)$.**

Table 7 D_4 equivariance error for unpooled and pooled observable-grid maps (lower is better; the pooled map is group-averaged over all eight square symmetries).

Model	Mean error	Std. error
D_4 -pooled PCAE2D	9.57×10^{-17}	1.26×10^{-17}
No positional encoding	7.40×10^{-1}	1.52×10^{-1}
PCAE2D positional	8.15×10^{-1}	1.29×10^{-1}
Local/raw	8.37×10^{-1}	1.31×10^{-1}

The numerical check below therefore tests the implementation, not the property. We run it over 7000 patch transforms of cached CIFAR images (Table 7, Figure 5). The pooled map’s equivariance error is $9.57 \times 10^{-17} \pm 1.26 \times 10^{-17}$, which is floating-point noise. The unpooled positional map has an error of 0.815 ± 0.129 , and the local/raw control has 0.837 ± 0.131 . Both are of order one. The patch grid’s geometry therefore makes the unpooled map D_4 -motivated only, and the explicit pooling construction is what delivers the property. This construction operates at the feature-grid level and serves here as a numerical diagnostic, rather than as a component of the trainable reconstruction pipeline. Integrating it end-to-end (Section 5.4) is concrete future work. **This route to symmetry differs from equivariant quantum neural networks, in which every circuit layer is built to commute with the group action (Nguyen et al. 2024). A recent image classifier of that kind builds C_4 rotation symmetry into the circuit itself (San Sebastian Sein et al. 2025). In PCAE the symmetry comes from averaging the observable grid over the group after measurement. The circuit itself is not equivariant, and the same average applies equally to a classical feature map. [Student: Table 7 labels the second column “Std. error”, but the text reads it as a standard deviation over 7000 transforms. Please confirm which it is and fix the header.]**

4.8 Entanglement Pair-observable necessity in the nonlocal-structure protocol

The diagnostic tests whether an-entangling a pair-observable channel is needed to fit a target that depends on pair correlations. The target is a fixed synthetic function of *simulated* patch features, and those features are not concatenated into any candidate’s own input block. The companion supplement gives the leakage-audit procedure that rules out circularity. Every control receives the same raw inputs as PCAE2D. **The task is by construction favourable to pair-observable structure, which is why we read it as a representational statement and not as a performance comparison.**

Table 8 and Figure 6 report the result across five seeds. PCAE2D’s entangling observable **pair-observable** channel reaches mean $R^2 = 0.9735$, or 32.77 dB PSNR. This is a 17.74 dB improvement over a no-entanglement control ($R^2 = 0.019$) and

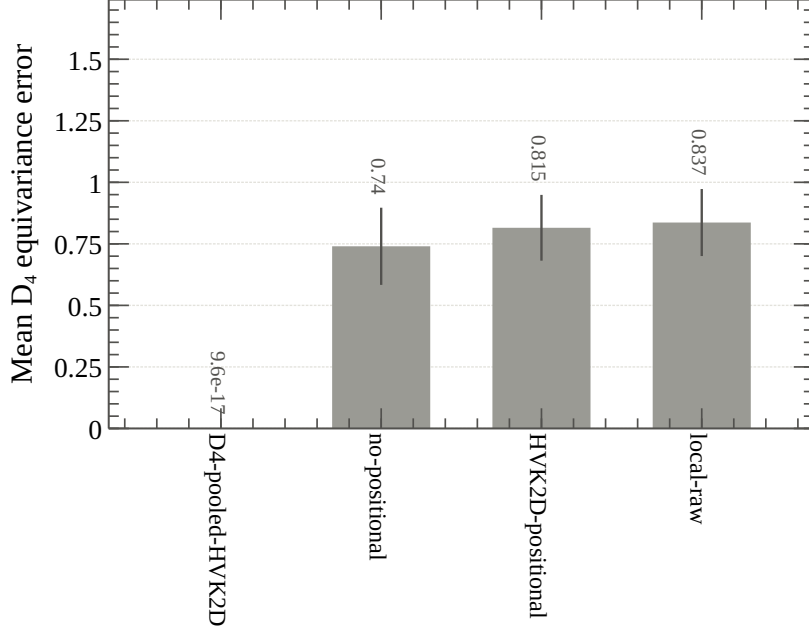


Fig. 5 D_4 equivariance diagnostic. The unpooled positional map is not equivariant; the group-pooled map is equivariant to numerical precision. The group-pooled map is equivariant to numerical precision. The unpooled positional map is not.

18.52 dB over random-VQC latents ($R^2 = -0.173$). Freezing the quantum circuit, so that features are fixed and only the readout is trained, still reaches $R^2 = 0.853$. A parameter-matched classical control ($R^2 = -0.030$) and a frozen-classical control ($R^2 = -0.903$) fail outright. **The raw-linear classical control ($R^2 = 0.0133$) is in the table but is not named here; whether it should be depends on the duplicate-control question raised separately. [Student: please state what the no-entanglement control measures. If it still measures ZZ on a product state, then $\langle Z_i Z_j \rangle = \langle Z_i \rangle \langle Z_j \rangle$ exactly, and its failure needs one sentence of explanation. If it drops the pair channel, its name should be “no pair observables”.]**

The mechanism is explicit under the fixed linear readout. A linear map of single-patch expectations $\{\langle O_i \rangle\}$ cannot form the cross-term $\langle O_i \rangle \langle O_j \rangle$ that the target depends on. The ~~entangling~~ **pair** measurement supplies the joint correlator $\langle O_i O_j \rangle$ directly as a feature. **Formally, let $z_i = \langle O_i \rangle$ and let every admissible predictor be affine, $f(z) = w^\top z + b$. Then $\partial^2 f / \partial z_i \partial z_j = 0$ for all i, j . A target with a term $c z_i z_j$, $c \neq 0$, has this mixed derivative equal to c wherever z_i and z_j vary independently, so no affine f can fit it. Adding $z_i z_j$ to the feature vector makes the term linearly representable.** ~~An entangling~~ **A** pair-observable channel is therefore *representationally necessary within the tested feature class and fixed linear-readout protocol* to fit this target. ~~A sufficiently nonlinear classical readout could in principle recover the same cross-terms, which is why we scope~~ **A classical**

Table 8 PCAE2D restricted pair-correlation diagnostic across five seeds. Deliberately favourable to pair-observable structure; evidence for ~~entanglement-necessity~~ **pair-observable necessity** under a matched linear readout.

Model	Mean MSE	Mean PSNR (dB)	Mean R^2
PCAE2D entangling pair observables	8.5969×10^{-4}	32.77	0.9735
Freeze quantum	4.7904×10^{-3}	27.34	0.8533
No entanglement	3.1461×10^{-2}	15.03	0.0191
Raw-linear classical	3.1654×10^{-2}	15.00	0.0133
Parameter-matched classical	3.3033×10^{-2}	14.82	-0.0297
Random VQC	3.7616×10^{-2}	14.25	-0.1732
Freeze classical	6.0973×10^{-2}	12.17	-0.9025

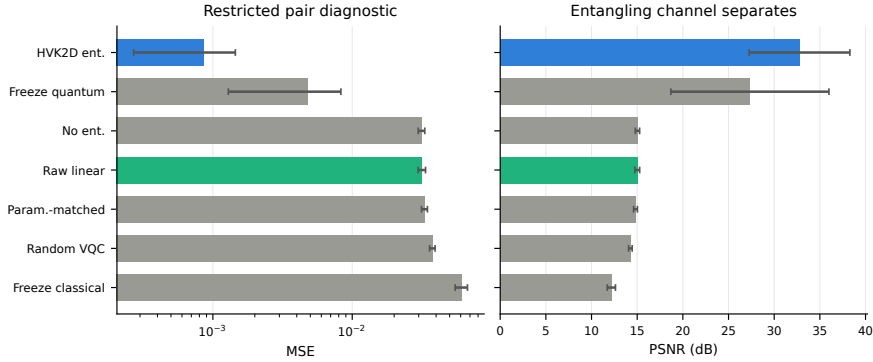


Fig. 6 Restricted pair-correlation diagnostic summary (mean $\pm$ std over five seeds). The ~~entangling observable~~ **pair-observable** feature map dominates every control on this deliberately favourable task.

model that forms the same products, for example through quadratic features or a nonlinear readout, would also fit this target. For this reason we scope the claim to the linear protocol rather than to classical models in general.

4.9 A scoped training-dynamics diagnostic

Beyond the results above, we tracked two exploratory training-dynamics readouts. These are a magnetisation-style statistic $M_z(t)$ and a signed energy-to-entanglement ratio $R_{ES}(t)$. Both drift smoothly over training with run-to-run fluctuations, and neither shows a sharp transition. Any apparent onset is a within-run median-plus- 2σ threshold crossing, rather than a calibrated statistical test.

A negative control fixes the scope. We applied the identical protocol to a classical-replacement variant, in which the Hamiltonian energy is identically zero throughout training. That variant crosses the threshold in all 4/4 tested cases, which is as often as the Hamiltonian-regularised model’s 4/4. The threshold therefore fires whether the Hamiltonian term is present or absent, and it does not track that term. ~~We make no~~

Table 9 Hamiltonian-sector controls (single-seed, fresh same-environment reproduction: *Monalisa*, 200 steps, seed 42). The energy objective is diagnostically useful but does not improve reconstruction accuracy. See the Supplementary Material for the full reproducibility investigation and additional observable-sector controls.

Experiment	PSNR (dB)	Interpretation
Legacy baseline, signed energy	38.63	Standard energy-regularised run
No energy loss	44.56	Removing energy improves PSNR
Contrastive Hamiltonian core	42.92	Stronger objective helps modestly vs. baseline, still below no-energy-loss

~~transition claim, and report~~ **We therefore report** these traces as an interpretability readout on the learned energy and entanglement. The companion Supplementary Material gives the full protocol and tables.

5 Discussion

5.1 ~~Why the physics-informed regulariser was inert~~ Effect of the Hamiltonian energy term

Table 9 shows the Hamiltonian energy term is a mild drag on reconstruction quality. The legacy objective adds a signed mean energy to the reconstruction loss. Because learned energy can go negative, the optimiser can lower total loss by driving energy down without improving reconstruction. Removing the term improves PSNR from 38.63 to 44.56 dB. A contrastive variant, assigning lower energy to correct feature-position pairs than to shuffled pairs, recovers part of the gap (42.92 dB) but stays below its no-energy counterpart. These are fresh same-environment measurements (*Monalisa*, 200 steps, seed 42), and they supersede earlier values produced by a script that cannot be audited.

The learned couplings enter only the energy term and never reach the decoder. The Hamiltonian is therefore a *read-only* observable of the trained representation rather than an inductive bias on reconstruction, and this is the sense in which we call it a diagnostic. A physically motivated regulariser has room to help when it resolves an otherwise underdetermined objective. This patch-level pixel task does not meet that condition, because the reconstruction loss already dominates the optimisation signal at this model scale. The Supplementary Material gives the full account. **[Student: two points in this section. (1) The text calls the Hamiltonian a read-only observable. But removing the energy term changes PSNR from 38.63 to 44.56 dB (Table 9), so its gradient does change the trained circuit, unless the energy is computed on detached features. Please reword “read-only”. (2) A gap of this size is more than “a mild drag”. Please choose wording that matches the table.]**

The training-dynamics readouts of Section 4.9 provide a descriptive window into how the learned energy and entanglement evolve. They are descriptive readouts rather than calibrated statistical tests, and the thermodynamic reading is outside their scope.

5.2 Task-dependent representational scope

Whether a quantum feature map offers any advantage over a classical model is a property of the specific data and target, not a fixed property of the model class. ~~Dequantization~~ **Dequantisation** results have repeatedly shown that apparent quantum advantages on structured, low-entanglement classical data can be matched once the right classical feature is identified (Huang et al. 2021; Tang 2019). Where provable separations do exist, they are established for carefully constructed learning problems rather than for generic structured data (Gyurik and Dunjko 2023). **We separate three questions. First, PCAE at six qubits can be simulated exactly on a classical computer, which says little about larger scales. Second, whether its model class admits an efficient classical replacement at scale is open. Tensor-network analyses give conditions for such dequantisation of variational models (Shin et al. 2024), and a classical tie at small scale does not settle it. Random Fourier features can also replace some variational models efficiently, under stated conditions (Sweke et al. 2025). Quantum convolutional networks on common image benchmarks can be simulated efficiently by a classical method based on Pauli shadows (Bermejo et al. 2026). Third, on the tested ordinary reconstruction tasks, PCAE stays within the declared 1 dB margin of the stronger matched classical controls. This matches large benchmarks in which simple classical models often match or beat small quantum models, and removing entanglement often costs little (Bowles et al. 2024). In PCAE the quantum processor evaluates only the circuit and returns the vector of measured Pauli expectation values. The MPS compression is quantum-inspired but runs on classical hardware, as do the Fourier positional encoding, the D_4 pooling, the decoder and the optimisation. Any quantum contribution to PCAE must therefore enter through the measured observable vector.**

The restricted diagnostic of Section 4.8 is constructed so that the relevant structure is a genuine distant-element product, which raw or local features cannot represent under a fixed linear readout. That is the regime in which PCAE’s ~~entangling~~ **pair-observable** channel separates. Natural image patches at the resolution and patch size used here are low-entanglement, locally structured data, where a linear or shallow classical readout is already close to sufficient. The distinction lies in the measured pair-observable channel rather than in the circuit. The ansatz uses strongly-entangling layers and does generate an entangled state, but whether the measured channel carries task-relevant information is a property of the target. On natural-image targets the pair observables add little beyond the local ones. On the constructed nonlocal target they are necessary. Together with the held-out comparisons of Section 4.3, these results delineate where the tested entangling representation does and does not yield a measurable benefit.

~~PCAE’s three ingredients each recur~~ **Three of PCAE’s ingredients recur** individually in the hybrid quantum-vision literature, covering MPS compression of structured input (Stoudenmire and Schwab 2016; Han et al. 2018; Cirac et al. 2021; Guala et al. 2023), a Pauli-correlator latent space handed to a classical decoder (Havlíček et al. 2019), and a physically motivated auxiliary regulariser alongside a task loss (Shi et al. 2023). Its specific instantiation, a six-qubit VQC with a 19–27-D correlator vector and a graph-Hamiltonian term, sits within the parameter ranges typical of that literature. The measured behaviour is therefore evidence about the design pattern itself. We do not claim the boundary transfers unchanged to every dataset, patch size, or qubit count.

5.3 Topology alignment as an inductive bias

Aligning the latent interaction graph with image-patch adjacency (PCAE2D vs. PCAE1D) is an architectural inductive bias, and the companion supplementary study tests it directly under matched held-out conditions. Its real-circuit subset (3 seeds, 3 held-out CIFAR-10 images) gives a PCAE1D-minus-PCAE2D mean PSNR difference of 0.16 dB. After averaging images within seed, the bootstrap 95% CI is $[-0.38, 0.46]$ dB and the seed-level Wilcoxon test gives $p = 0.50$. A broader five-seed surrogate comparison across CIFAR-10, Fashion-MNIST, and PathMNIST likewise finds numerically small topology differences, at most 0.07 dB in mean PSNR. PCAE2D completes the matched 90-step runs in roughly half the wall-clock time. At the tested scale, topology is therefore primarily a resource-cost choice rather than a reliable accuracy advantage. These tests do not establish formal equivalence.

Table 10 contrasts PCAE’s structural properties against reference frameworks, including the three nearest hybrid-vision architecture families. Its distinguishing features are a Pauli-correlator latent space, a 2D Fourier positional bias with enforced D_4 pooling, and a Hamiltonian energy diagnostic, each documented and scoped in the preceding sections.

5.4 Validated scope

- ~~Every reconstruction result in this paper (Sections 4.1–4.2) is per-image trained. The companion supplementary study’s held-out protocol is the appropriate evidence for a generalisation claim.~~ **Every reconstruction result in this paper is per-image trained, except the held-out comparison of Section 4.3. Only that held-out comparison, reported in full in the companion supplement, supports a generalisation claim.**
- The D_4 -pooled map (Section 4.7) is a feature-grid-level construction and numerical diagnostic. Integrating it into the trainable end-to-end pipeline is future work (Section 6).
- ~~The entanglement-necessity~~ **pair-observable necessity** result (Section 4.8) is scoped to a task constructed to require nonlocal structure. Ordinary natural-image reconstruction lies outside that scope (see Section 5.2).

Table 10 Architectural differentiation from representative baseline designs, including the three nearest hybrid-vision architecture families (register-based QAE, quantum-kernel vision, TN-circuit classifier).

Axis	Conv. AE baseline	GAN baseline	Register-based QAE	Quantum-kernel vision	TN-circuit classifier	PCAE (ours)
Latent representation	Continuous $\mathbb{R}^d$	Continuous $\mathbb{R}^d$	Quantum register	Scalar kernel value	MPS/circuit state	Pauli-correlator vector
Spatial bias	Convolutional	Convolutional	Encoding-dependent	Encoding-dependent	MPS chain ordering	2D Fourier encoding
Auxiliary objective	Bottleneck / reconstruction	Adversarial loss	Fidelity / reconstruction	None (kernel value only)	Classification loss	Hamiltonian diagnostic
Enforced group symmetry	Not in baseline	Not in baseline	Not in reference design	Not in reference design	Not in reference design	D_4 observable pooling
Evaluated role	Reconstruction	Reconstruction	Compression	Classification	Classification	Reconstruction + diagnostics

- The hardware pilot (Section 4.4) is a five-image, ~~single-trial~~, unmitigated pilot **with one run per image on the main backend** rather than a statistically resolved hardware benchmark.
- The training-dynamics traces (Section 4.9) are an interpretability readout. The threshold fires whether the Hamiltonian term is present or absent, at 4/4 in both cases, with the full table in the Supplementary Material.
- The matched real-circuit topology comparison uses 3 seeds and 3 held-out images per topology, and the broader five-seed comparison is surrogate-based. These results support a resource-cost interpretation rather than formal equivalence of the topologies.
- Every result here is at a six-qubit, depth-18 scale. Trainability of variational models is known to degrade with width and depth through barren-plateau effects (McClean et al. 2018; Larocca et al. 2025), so whether the tested boundary moves at larger scale is an open question.

6 Conclusion

We presented the ~~Hamiltonian-Vision-Kernel~~ **Pauli-Correlator Autoencoder (PCAE)**, a hybrid quantum-classical architecture for image reconstruction that assembles matrix-product-state patch features, a measured Pauli-correlator latent space, enforced D_4 -equivariant observable pooling, and a learnable Hamiltonian energy diagnostic into a single pipeline (Section 5.2). Across six real image datasets, per-image-trained PCAE2D models ~~reconstruct with high fidelity~~ **reach 25.8–41.6 dB**

PSNR. On a resource-matched, multi-seed held-out comparison, a pre-declared TOST equivalence test shows PCAE2D is *competitive with*, rather than inferior or superior to, simple classical controls (Section 4.3). A hardware pilot replayed trained checkpoints on IBM Quantum devices without retraining, establishing feasibility at a five-image, single-trial scope **with one run per image on the main backend** (Section 4.4).

HVK’s case rests on capabilities a classical reconstruction pipeline does not provide. Pair observables are necessary on a task requiring nonlocal structure, ~~within the tested fixed-readout feature class~~ **under a fixed linear readout** (Section 4.8). The Hamiltonian energy term serves as an interpretable diagnostic of the trained representation (Section 5.1). The pipeline executes natively on real quantum hardware. It is also exactly D_4 -equivariant by construction, which we report as a design-correctness property rather than a quantum capability, since the same Reynolds averaging applied to a classical feature map is equally exact. We make no claim of quantum advantage. ~~HVK is, at this scope, a new architecture that works, is competitive with resource-matched classical baselines, and runs on real quantum hardware.~~ **At the six-qubit scale tested here, PCAE is competitive with resource-matched classical baselines and runs on real quantum hardware.**

Future work should integrate the D_4 -pooled map into an end-to-end trainable pipeline (Section 5.4), scale the hardware pilot toward a statistically resolved benchmark with error mitigation and repeated trials, search for further real-world tasks with nonlocal structure where PCAE’s ~~entanglement-sensitive~~ **pair-observable** separation translates into a reconstruction advantage, and extend the resource-matched, leakage-audited evaluation protocol to other hybrid quantum vision architectures.

Supplementary information. A companion document, *Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility*, accompanies this manuscript.

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Statements and Declarations

Competing interests. The authors declare no competing interests.

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Data availability. Dataset provenance and access routes are cited in the manuscript and detailed in the supplementary artifact map. All datasets used (CIFAR-10, MNIST, Fashion-MNIST, and the PathMNIST, BloodMNIST and PneumoniaMNIST subsets of MedMNIST v2) are publicly available.

Code availability. The source code, experiment drivers, machine-readable result summaries, validation arrays, and figure inputs supporting this study are included in the accompanying repository. The file `project_artifacts/submission_claim_audit.json` maps each numerical headline claim to its retained source artifact, covering the resource-matched held-out comparison, on which the tested PCAE map is competitive with rather than superior

to the classical controls, and the withdrawn training-dynamics material. **[Student: Section 4.9 still reports the training-dynamics traces as an interpretability readout. Please say which training-dynamics material was withdrawn, so that the two statements agree.]** Automated tests verify those mappings. IBM Quantum job identifiers and reconstructed outputs are retained; as disclosed in the supplement, the raw account-usage service response was not serialised, so the reported quota change is an execution-log record rather than a repository-recomputable measurement.

Author contributions. **Sparsho Chakraborty:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Siddhartha Patra:** Conceptualization, Supervision, Writing – review & editing, Resources. Both authors reviewed and approved the final manuscript.

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